

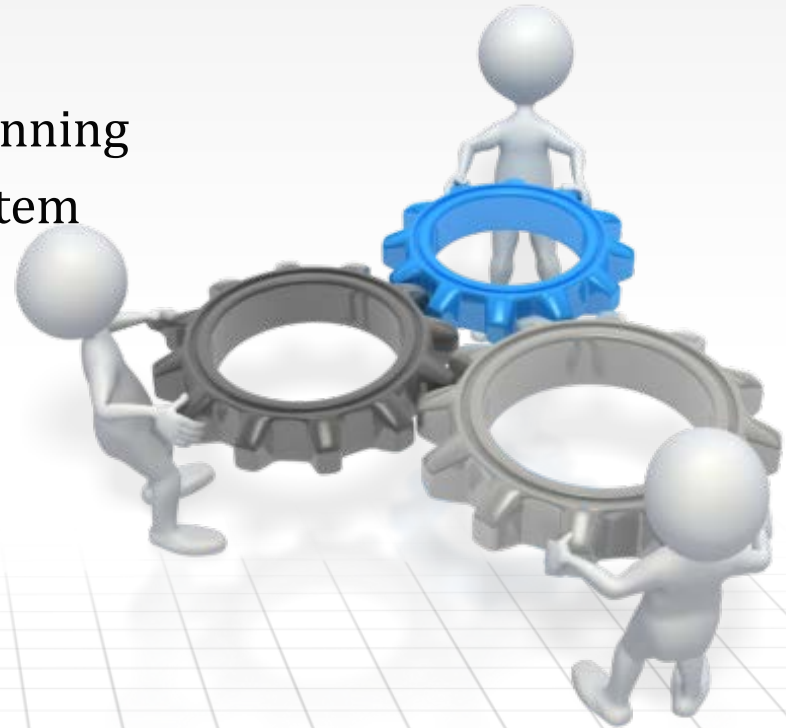


LECTURE 10

PRODUCTION CONTROL SYSTEMS

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4. Inventory control
5. Manufacturing Resource Planning
6. Just-in-Time production system
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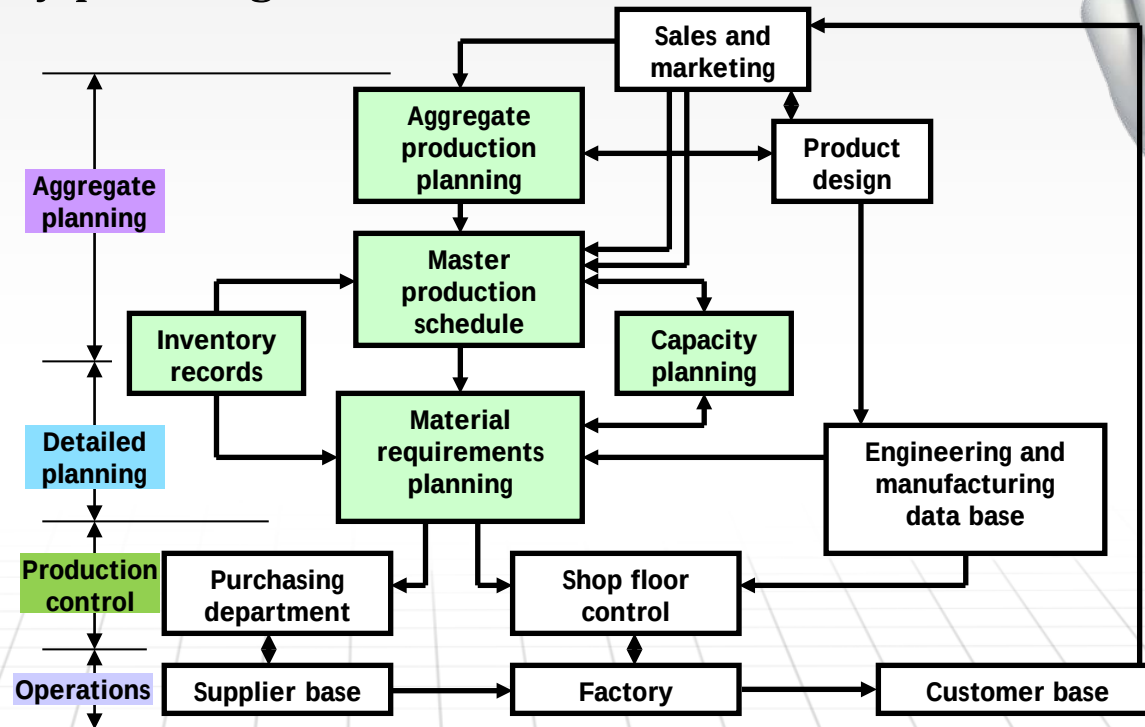
Objectives

1. Knowing the production planning strategies
2. Knowing the management of material requirements
3. Understanding the allocation of manufacturing resources
4. Identifying the specific features of Just-in-Time manufacturing systems



Review

1. Aggregate production planning
2. Master production planning
3. Material requirements planning (MRP)
4. Capacity planning

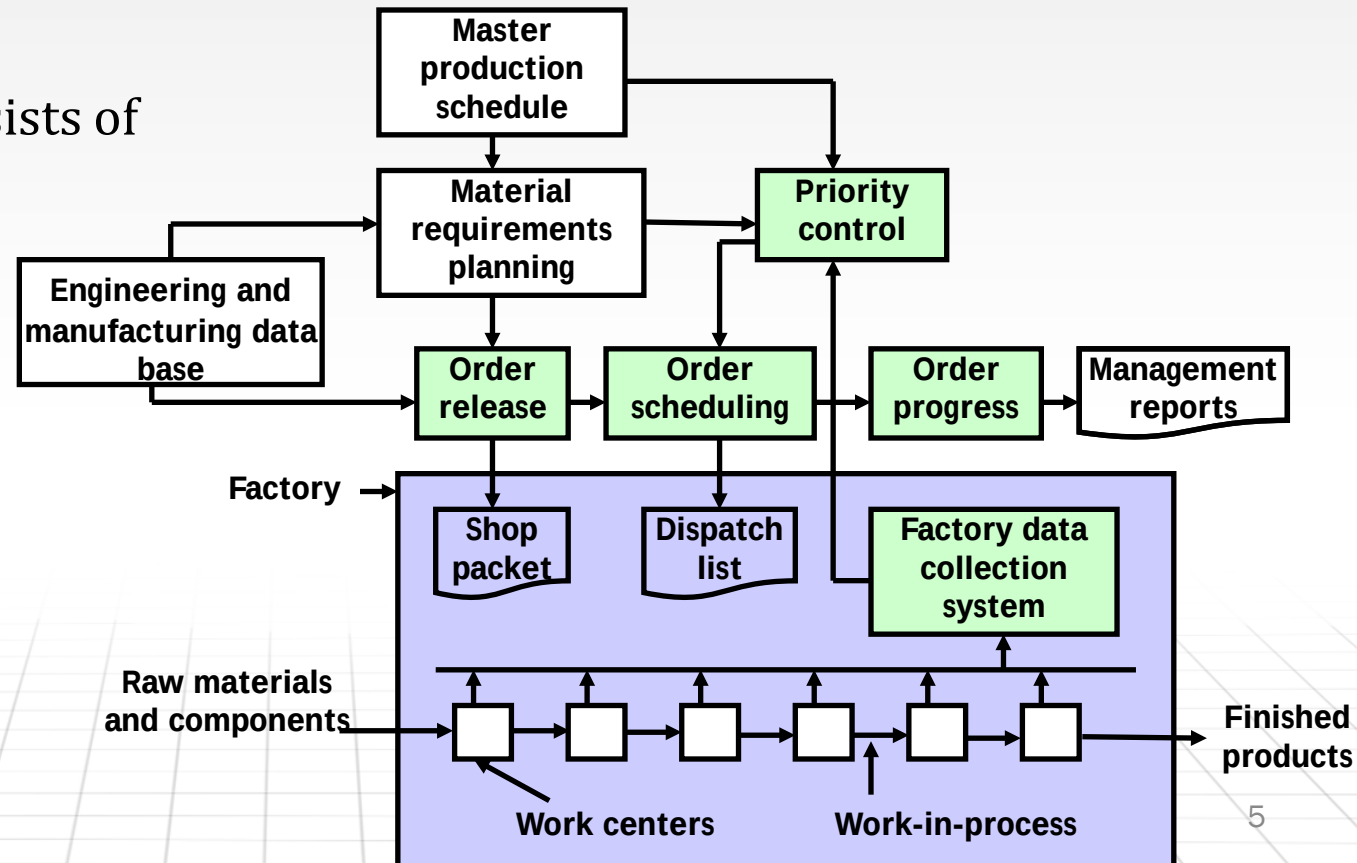


Shop floor control

Shop floor control is concerned with the release of production orders to the factory, monitoring and controlling the progress of the orders through the various work centers, and acquiring current information on the status of the orders.

□ A typical shop floor control system consists of three phases:

- order release,
- order scheduling,
- order progress.





Order release

Order release provides the documentation needed to process a production order through the factory.

❑ Order release consists of:

- *route sheet* – documents the process plan for the item to be produced,
- *material requisitions* – to draw the necessary raw materials from inventory,
- *job cards* – to report direct labor time devoted to the order and to indicate progress of the order through the factory,
- *move tickets* – to authorize the material handling personnel to transport parts between work centers,
- *parts list* – if required for assembly jobs.



Order scheduling

Order scheduling assigns the production orders to the various work centers in the plant.

- ❑ The order scheduling module:
 - prepares a *dispatch list* which indicates which production order should be accomplished at the various work centers.
 - provides information about relative priorities of the different jobs .
- ❑ Some of the dispatching rules used to establish priorities for production orders in the plant include:
 - shortest operation next
 - first come-first served
 - critical ratio rule
 - order slack time rule
 - slack time per remaining operations rule
 - queue ratio rule
 - earliest due date rule



**Shortest
operation next**

The order with the shortest operation time has the highest priority.

Example 1

Calculate the priority schedule for the shop orders shown below, using the shortest operation next rule.

Shop order number	1A	2A	3A
Data arrived	117	115	116
Date due	125	120	130
Lead time remaining (LTR)	8	5	14
Processing time remaining (PTR)	6	4	5
Operations remaining (OR)	1	2	8
Standard scheduled queue (SSQ)	2	1	5
Next operation time	6	3	1

All Times in Days; Today's Date, M-Day 117

Solution:

Priority	Order number	Next operation time
1	3A	1
2	2A	3
3	1A	6

This rule maximizes the number of shop orders processed and minimizes the number waiting in queue.



First come-first served

The priority is set by the arrival date of the shop order in the work center.

Example 2

Calculate the priority schedule for the shop orders shown below, using the first come-first served rule.

Shop order number	1A	2A	3A
Data arrived	117	115	116
Date due	125	120	130
Lead time remaining (LTR)	8	5	14
Processing time remaining (PTR)	6	4	5
Operations remaining (OR)	1	2	8
Standard scheduled queue (SSQ)	2	1	5
Next operation time	6	3	1

All Times in Days; Today's Date, M-Day 117

Solution:

Priority	Order number	Arrival date
1	2A	115
2	3A	116
3	1A	117



The rule schedules on the basis of the arrival time of the part in the work center and will result in the shortest average queue times for parts.

Critical ratio (CR)

The rule is based on the ratio of time remaining to work remaining.

$$CR = \frac{\text{due date} - \text{today's date}}{\text{lead time remaining}}$$

Example 3

Calculate the priority schedule for the shop orders shown below, using the critical ratio rule.

Shop order number	1A	2A	3A
Data arrived	117	115	116
Date due	125	120	130
Lead time remaining (LTR)	8	5	10
Processing time remaining (PTR)	6	4	5
Operations remaining (OR)	1	2	8
Standard scheduled queue (SSQ)	2	1	5
Next operation time	6	3	1

All Times in Days; Today's Date, M-Day 117

Solution:

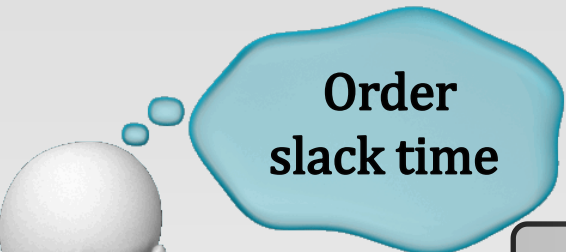
$$\text{Order 1A: } CR = \frac{125 - 117}{8} = \frac{8}{8} = 1 \text{ (on schedule)}$$

$$\text{Order 2A: } CR = \frac{120 - 117}{5} = \frac{3}{5} = 0.6 \text{ (behind schedule)}$$

$$\text{Order 3A: } CR = \frac{130 - 117}{10} = \frac{13}{10} = 1.3 \text{ (ahead of schedule)}$$

Priority	Order number	Critical ratio
1	2A	0.6
2	1A	1.0
3	3A	1.3

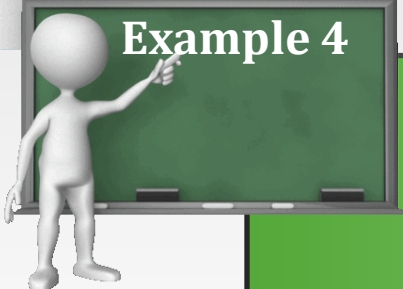
The rank order of the jobs by ratio makes sure that those behind schedule are performed first.



Order slack time

The slack formula is as follows:

Slack = due date – today's date – processing time remaining



Example 4

Calculate the priority schedule for the shop orders shown below, using the order slack rule.

Shop order number	1A	2A	3A
Data arrived	117	115	116
Date due	125	120	130
Lead time remaining (LTR)	8	5	10
Processing time remaining (PTR)	6	4	5
Operations remaining (OR)	1	2	8
Standard scheduled queue (SSQ)	2	1	5
Next operation time	6	3	1

All Times in Days; Today's Date, M-Day 117

Solution:

Order 1A: Slack = 125 – 117 – 6 = 2 (ahead of schedule)

Order 2A: Slack = 120 – 117 – 4 = –1 (behind schedule)

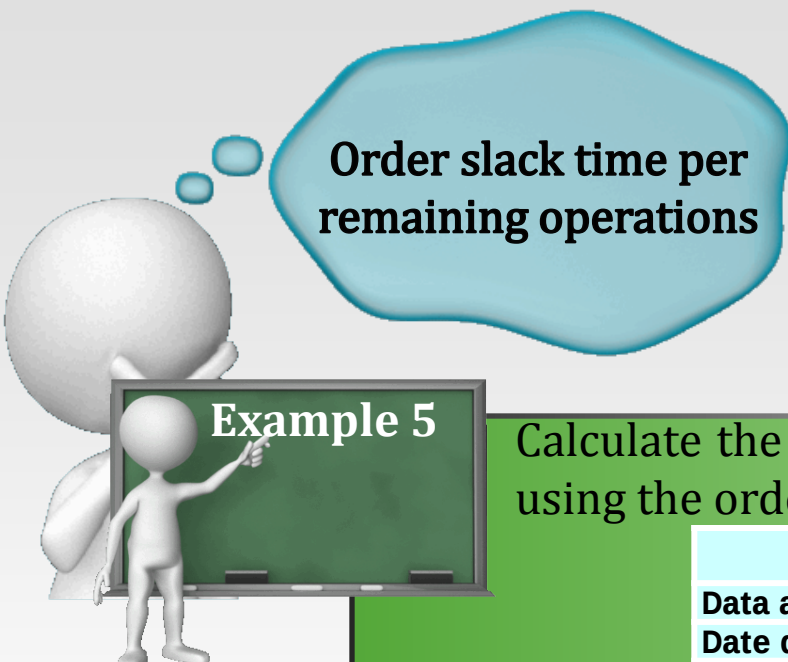
Order 3A: Slack = 130 – 117 – 5 = 8 (ahead of schedule)

Priority	Order number	Slack
1	2A	-1
2	1A	2
3	3A	8



This priority rule addresses the sequencing of jobs on the basis of the value of work remaining.

*Slack time is the total time a task can be delayed without causing additional delays



Order slack time per remaining operations

The rule is based on the ratio:

$$\text{Slack per operations (S/O)} = \frac{\text{slack time}}{\text{operations remaining}}$$

Example 5

Calculate the priority schedule for the shop orders shown below, using the order slack time per remaining operations rule.

Shop order number	1A	2A	3A
Data arrived	117	115	116
Date due	125	120	130
Lead time remaining (LTR)	8	5	10
Processing time remaining (PTR)	6	4	5
Operations remaining (OR)	1	2	8
Standard scheduled queue (SSQ)	2	1	5
Next operation time	6	3	1

All Times in Days; Today's Date, M-Day 117

Solution:

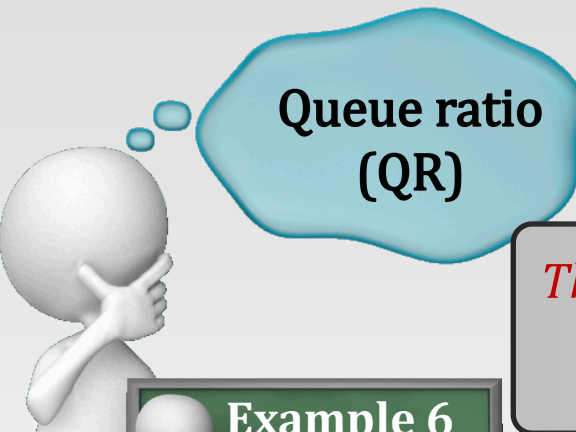
Order 1A: $S/O = \frac{2}{1} = 2.0$ (ahead of schedule)

Order 2A: $S/O = \frac{-1}{2} = -0.5$ (behind schedule)

Order 3A: $S/O = \frac{8}{8} = 1.0$ (on schedule)

Observation: Values for slack time are obtained from Example 4

Priority	Order number	Slack
1	2A	-0.5
2	3A	1.0
3	1A	2.0



Queue ratio
(QR)

The rule is based on the formula:

$$QR = \frac{\text{slack time}}{\text{standard scheduled queue}}$$

Example 6

Calculate the priority schedule for the shop orders shown below:

Shop order number	1A	2A	3A
Data arrived	117	115	116
Date due	125	120	130
Lead time remaining (LTR)	8	5	10
Processing time remaining (PTR)	6	4	5
Operations remaining (OR)	1	2	8
Standard scheduled queue (SSQ)	2	1	5
Next operation time	6	3	1

All Times in Days; Today's Date, M-Day 117

Solution:

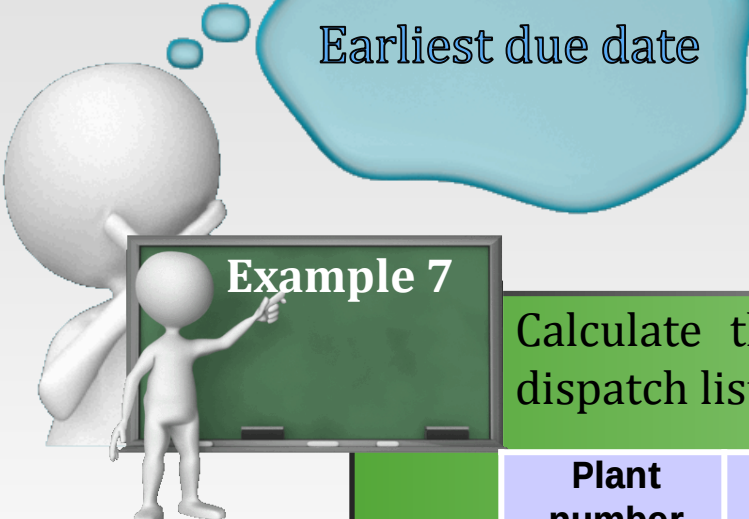
Order 1A: $QR = \frac{2}{2} = 1.0$ (on schedule)

Order 2A: $QR = \frac{-1}{1} = -1.0$ (behind schedule)

Order 3A: $QR = \frac{8}{5} = 1.6$ (ahead of schedule)

Observation: Values for slack time are obtained from Example 4.

Priority	Order number	Slack
1	2A	-1.0
2	1A	1.0
3	3A	1.6



Earliest due date

The rule schedules order priority according to the order due date

Example 7

Calculate the priority schedule for the shop orders on the dispatch list shown below:

Plant number	Order number	Quantity	Hr/Unit	Total hours	Due date
12-9201	SO 73421	300	0.2	60	9-20
12-4565	SO 73107	100	0.3	30	9-22
18-5632	SO 73560	150	0.2	30	9-28
				120	

Solution:

Priority	Order number
1	SO 73421
2	SO 73107
3	SO 73560



Self-assessment test

1. Order release module may contains the list of necessary raw materials.

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2. Order scheduling module provides information about the priority order of operations.

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3. Shortest operation next rule maximize the number of processed orders.

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4. Order slack time rule addresses the sequencing of jobs on the basis of the value of work remaining.

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5. When earliest due date rule is used, the highest priority is assigned to the part with the latest due date.

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Order progress

The order progress module monitors the status of the various orders in the plant, work-in-process, and other characteristics that indicate the progress and performance of production.

- ❑ The information presented to production management is often summarized in the form of reports like:
 - work order status reports
 - progress reports
 - exception reports
- ❑ These reports are useful to production management in:
 - making decisions about allocation of resources;
 - authorization of overtime hours;
 - identifying problem areas in the plant that adversely affect achieving the MPS.



Factory data collection system

The factory data collection system consists of the various paper documents, terminals, and automated devices located throughout the plant for collecting data on shop floor operations, plus the means for compiling and processing the data.

- ❑ The purpose of the factory data collection system is:
 - to supply status and performance data to the shop floor control system;
 - to provide current information to production foremen, plant management, and production control personnel.



Inventory control (1)



- ❑ The types of inventory of greatest interest in PPC are:
 - raw materials,
 - purchased components,
 - in-process inventory (WIP),
 - finished products.
- ❑ The major costs of holding inventory are:
 - investment costs,
 - storage costs,
 - cost of possible obsolescence or spoilage



Inventory control (2)



❑ Inventory control procedures:

- for dependent demand – MRP is the most widely implemented technique;
- for independent demand – order point inventory systems are used.

❑ Order points systems are concerned with two related problems that must be solved when managing inventories of independent demand items:

- how many units should be ordered? -> economic order quantity formulas,
- when should the order be placed? -> record point methods



Manufacturing resource planning (MRP II) (1)

Manufacturing resource planning can be defined as a computer based system for planning, scheduling, and controlling the materials, resources, and supporting activities needed to meet the MPS.

- ❑ Application modules provided in a high-end MRP II system:
 - *management planning* (business strategy, aggregate production planning, MPS, budget planning)
 - *customer service* (sales forecasting, order entry, sales analysis, finished goods inventory)
 - *operations planning* (MRP+ capacity requirements planning)
 - *operations execution* (purchasing, production scheduling and control, WIP inventory control, shop floor control, labor hour tracking)
 - *financial functions* (cost accounting, accounts receivable, accounts payable)



Manufacturing resource planning (MRP II) (2)

❑ Benefits from MRP II:

- *Reduction in inventory* => less money spent on investments for material and the cost of keeping that material on the shelf until it is needed.
- *Better planning in communications* => reductions in the amount of inventory declared obsolete and written off as a loss.
- *Better scheduling* => allows better planning with suppliers and lower purchasing costs.
- *Customer service levels* are improved by shipping more complete orders on time and with improved responsiveness to customer requirements.

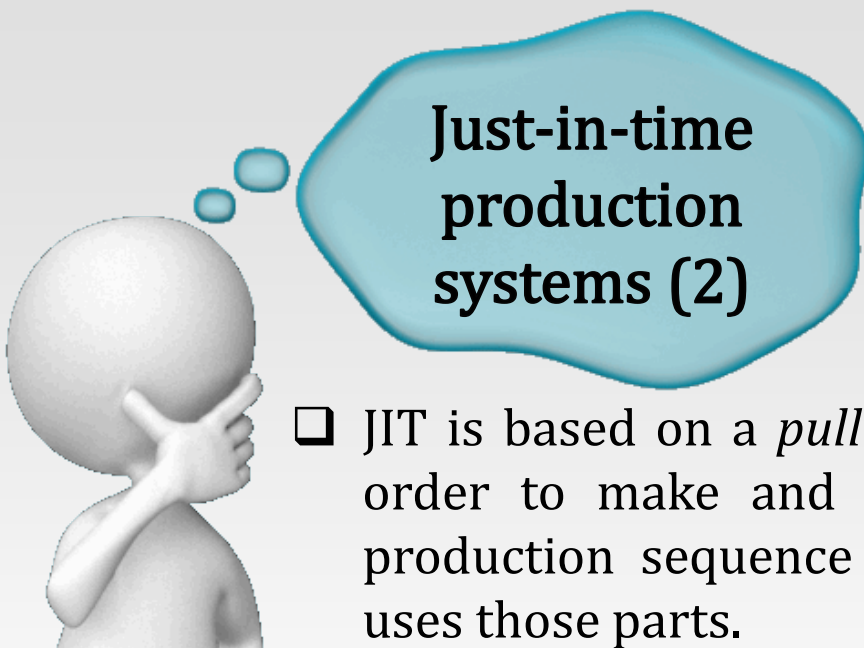


Just-in-time production systems (1)

Just-in-time production systems (JIT) were developed in Japan to minimize inventories, especially WIP.

- ❑ The ideal just-in-time production system produces and delivers exactly the required number of each component to the downstream operation in the manufacturing sequence just at the time when that component is needed.
- ❑ JIT operate successfully if we have:
 - a pull system of production control
 - small batch sizes
 - reduced setup times
 - stable and reliable production operations





Just-in-time production systems (2)

- ❑ JIT is based on a *pull system* of production control, in which the order to make and deliver parts at each workstation in the production sequence comes from the downstream station that uses those parts.
- ❑ One way to implement a pull system is to use kanbans. (kanban means card in Japanese).
- ❑ The Kanban system of production control, developed by Toyota, is based on the use of two types of cards that authorize parts production, and parts delivery in the plant:
 - production kanbans (b)
 - transport kanbans (a)

Part number _____			Preceding process _____
Part name _____			
Box capacity	Box type	Issue no.	Subsequent process _____

(a)

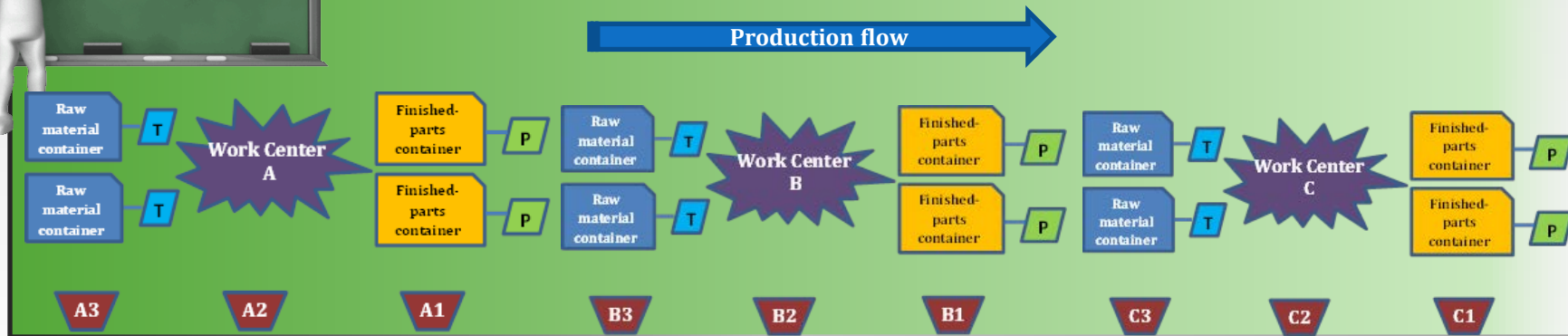
Part number _____	Process _____
Part name _____	
Stock location at which to store: _____	
Container capacity: _____	

(b)

Just-in-time production systems (3)

Example 8

A Kanban control system is illustrated in the figure below:



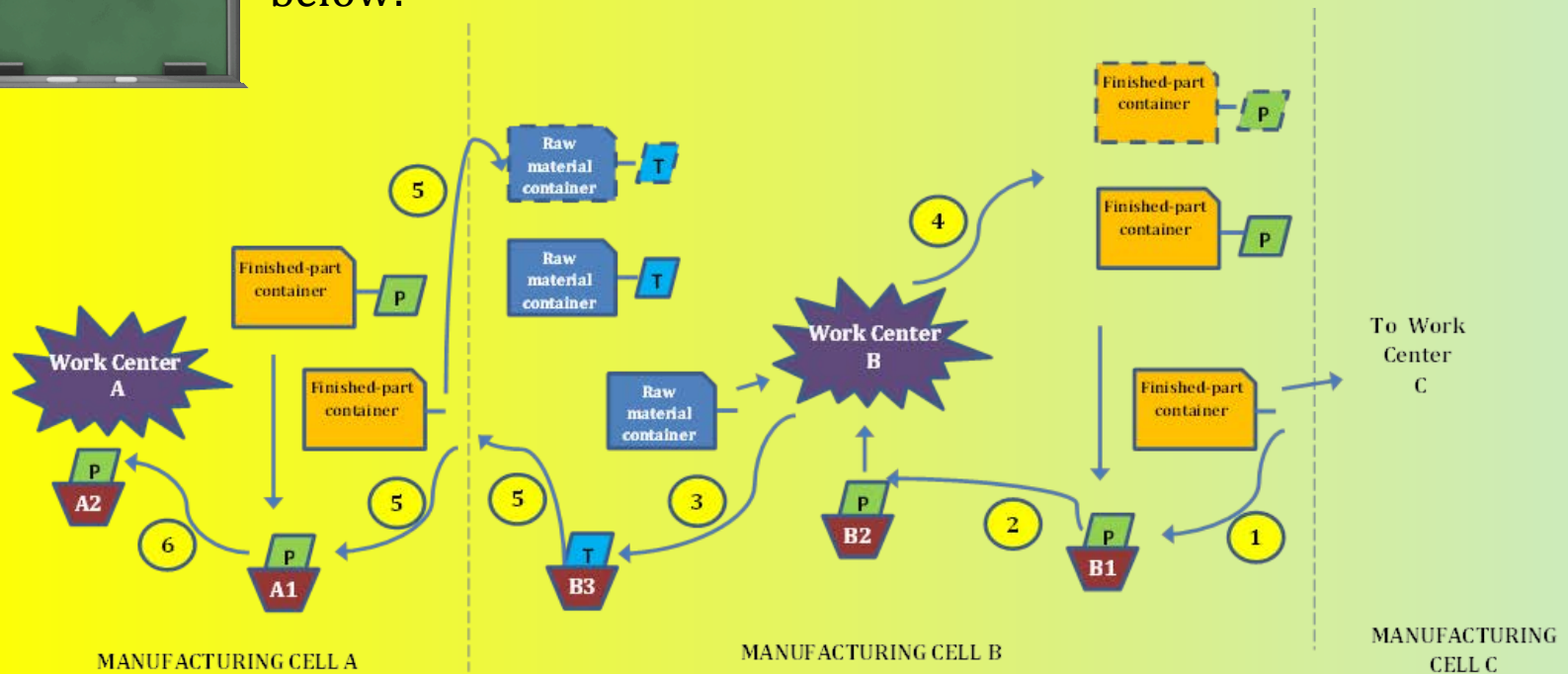
-  transport card
-  production card
-  Kanban cards boxes

- ☐ The production process begins with the moving of a finished-part container from Work Center B to Work Center C where the parts are used for assembly.

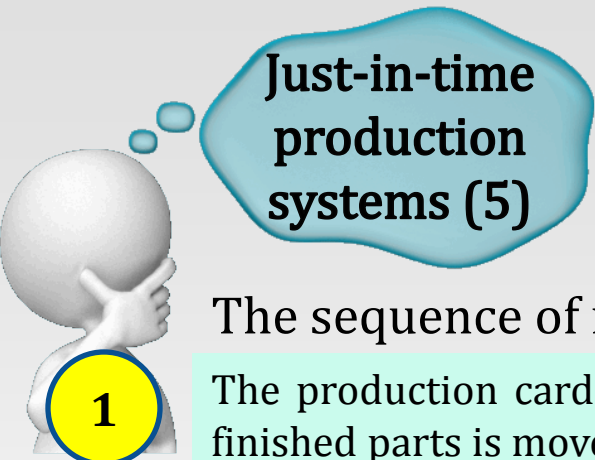
Just-in-time production systems (4)

Example 8

Changes that occur in work centers are illustrated in the figure below:



The sequence of movement of the cards is illustrated by arrows and numbers, and is explained on the next slide.²⁵



Just-in-time production systems (5)

The sequence of movement of the cards:

1

The production card is removed from the production container before the container with finished parts is moved from cell B to cell C. The card is placed in Kanban box B1.

2

An inventory specialist moves the production card from kanban box B1 to the work-cell kanban box B2. The arrival of a production card in B2 authorizes the work-cell operator to produce another container of finished parts at work-center B, to replace the container just removed (moved to C). The production card from B2 is retained by the operator at cell B.

3

Production is started by moving a container of raw materials into cell B from the input side. The transport, or conveyance, card in the raw material container is removed and put into Kanban box B3.

4

The operator in cell B finishes production on the container of parts and moves the finished-parts container to the output side of the cell B. The production card held by the operator (at the step 2) is placed on the finished container.

5

The inventory specialist retrieves the transport card from Kanban box B3. The card authorizes the movement of a finished-part container from the output side of cell A to the input side of cell B. The production card in the container is placed in Kanban box A1, and the transport card is placed in the container on the input side of cell B.



Self-assessment test

1. The order progress module generates reports on progress queues.

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2. The factory data collection system communicates with the control system.

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3. In order to restore stocks at certain points in time it is necessary to know the quantity to order.

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4. Operations planning is not for the manufacturing resource planner.

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5. In the case of "Just-in-Time" manufacturing systems, inventories represented by WIP are zero.

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Conclusion

- ❑ Orders scheduling is a particularly important activity because it determines the efficiency of production and ensure timely delivery of orders.
- ❑ Effective control of inventories provides cost reductions associated with them.
- ❑ Just-in-Time method ensures minimization of inventories, especially those due to the components in the production process.



- ☐ Quality in manufacturing
- ☐ Traditional an modern quality control
- ☐ Quality function deployment
- ☐ House of quality